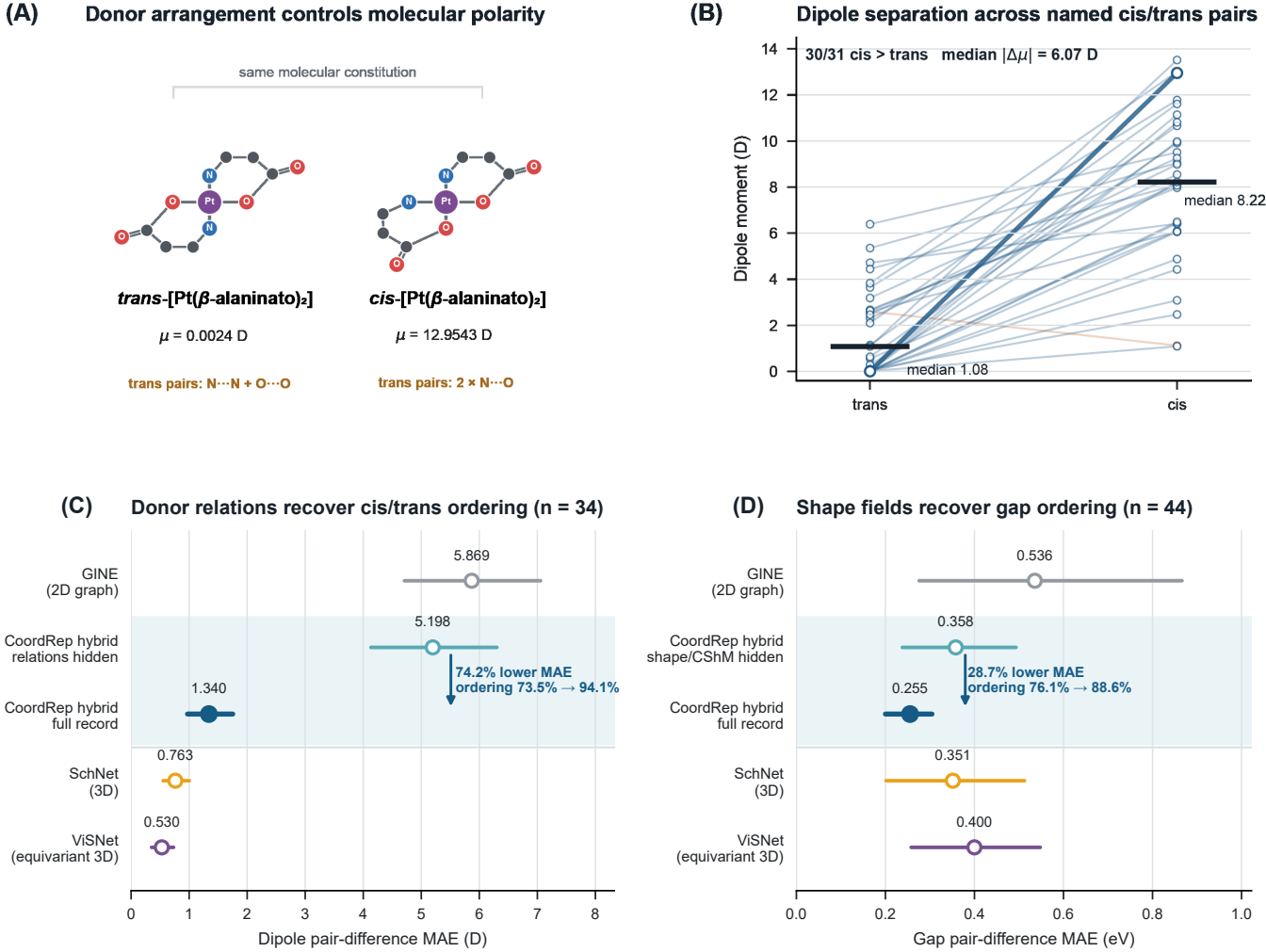

(A)
Donor arrangement controls molecular polarity
same molecular constitution
trans-[Pt(β-alaninato)₂]
μ = 0.0024 D
trans pairs: N···N + O···O
cis-[Pt(β-alaninato)₂]
μ = 12.9543 D
trans pairs: 2 × N···O
(B)
Dipole separation across named cis/trans pairs
30/31 cis > trans median |Δμ| = 6.07 D
Dipole moment (D)
median 8.22
median 1.08
trans
cis
(C)
Donor relations recover cis/trans ordering (n = 34)
GINE (2D graph)
5.869
CoordRep hybrid relations hidden
5.198
74.2% lower MAE
ordering 73.5% → 94.1%
CoordRep hybrid full record
1.340
SchNet (3D)
0.763
ViSNet (equivariant 3D)
0.530
Dipole pair-difference MAE (D)
(D)
Shape fields recover gap ordering (n = 44)
GINE (2D graph)
0.536
CoordRep hybrid shape/CShM hidden
0.358
28.7% lower MAE
ordering 76.1% → 88.6%
CoordRep hybrid full record
0.255
SchNet (3D)
0.351
ViSNet (equivariant 3D)
0.400
Gap pair-difference MAE (eV)